Scott Nguyen

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EDUCATION

Master of Science in Electrical Engineering | Space Systems Engineering **Spring 2026**
University of New Mexico GPA: 3.94/4.00
Master of Science in Aerospace Engineering **Fall 2024**
University of Illinois Urbana–Champaign
Bachelor of Science in Aerospace Engineering **Spring 2022**
Iowa State University

SKILLS

Programming Languages: MATLAB, Python, C/C++, Ruby
Frameworks / Libraries / Tools: NumPy, SciPy, Matplotlib, Astropy, Poliastro, bpy, git
Applications: Simulink, Blender, GRASP, SystemVue, Advanced Design System

WORK EXPERIENCE

Student Co-Op, Electronics for Contested Space Group **September 2025 – Present**
MIT Lincoln Laboratory
- Led design of a new **S-band recontact architecture** for a U.S. radar network, integrating flight-like hardware and simulation to reestablish links with lost-comm satellites.
- Implemented an ***Unscented Kalman Filter (UKF)*** for precise radio frequency measurements and state estimation
- Built a probabilistic detection tool to compute observation likelihoods from resident space object (RSO) properties and optical sensor performance.

Guidance, Navigation & Controls Engineer Intern **May 2025 – August 2025**
Blue Canyon Technologies
- Verified functionality and polarity of IMU, Nano Star Tracker, Reaction Wheels, Torque Rods, and Sun Sensors via hardware testing and data checks
- Performed regression analysis of two-axis ***Solar Array Drive Assembly (SADA)*** momentum management and validated command interfaces for precise control and reliability
- Automated ***SADA*** validation by developing ***Ruby*** test scripts and mapping telemetry channels to ***COSMOS***

Guidance, Navigation & Controls Engineer Intern **January 2025 – April 2025**
Blue Origin
- Integrated ***Active Disturbance Rejection Control (ADRC)*** and ***Sliding Mode Control (SMC)*** to develop a robust algorithm for stabilizing nonlinear MIMO dynamics of the BE-7 engine
- Evaluated control performance by injecting disturbances and demonstrated improved accuracy in setpoint tracking
- Integrated flight software into ***Simulink*** using ***S-functions*** in ***C*** to enable testing and verification

Guidance, Navigation & Controls Engineer Intern **May 2024 – August 2024**
Varda Space Industries
- Conducted trade studies to optimize gravity models for mission requirements and select optimal filter type (***EKF*** vs. ***UKF***)
- Built Monte Carlo simulations to quantify reentry uncertainty, generating latitude/longitude covariance ellipsoids and a reentry dispersion cloud for flight safety analysis and capsule recovery planning
- Implemented an ***EKF*** for state estimation, optimizing ground station timing for minimal residuals and precise delta-v planning
- Added unit tests and CI/CD pipelines using Bamboo for continuous integration

Guidance, Navigation & Controls Engineer Intern **May 2023 – August 2023**
Space Dynamics Laboratory
- Implemented a UKF with range iteration and least-squares orbit determination methods using optical navigation
- Performed Monte Carlo analysis on relative orbits to identify challenging scenarios and refine estimation algorithms
- Developed unit tests for ***Lambert Solver*** integrated with ***Initial Orbit Determination (IOD)*** pipeline

RESEARCH PROJECTS

Global Trajectory Optimization Competition (GTOC 6 & 11) – Modeling & Simulation

- Developed high-fidelity simulations for interplanetary trajectory design and optimization problems under nonlinear dynamics
- Contributed to solution verification and analysis workflows; maintained analysis notebooks and documentation for team reproducibility

Delta-V Minimization from Geostationary Orbit to Mars

- Applied trajectory optimization techniques to minimize delta-v for an Earth-to-Mars transfer, improving fuel efficiency
- Generated pork-chop plots using Lambert solutions and validated optimizer results against global minima
- Visualized optimized trajectories and planetary motion via Blender's Python API

TEACHING & MENTORING EXPERIENCE

Youth Development Professional — **October 2024 – January 2025**

Boys & Girls Club

- Designed and delivered an after-school **computer literacy** curriculum for grades 3–6 (typing, file systems, internet safety).
- Taught **introductory coding** via Scratch and beginner Python with project-based mini-units.
- Used quick pre/post checks and differentiated supports to boost engagement and skill growth.

Math Teacher (Algebra I, Algebra II, Precalculus, Calculus) — **October 2024 – January 2025**

North Star School

- Developed **unit plans, assessments, and problem sets** aligned to course objectives; integrated Desmos/GeoGebra.
- Differentiated instruction and provided timely, standards-based feedback.
- Connected math to **real-world STEM** applications to deepen understanding.

Research Assistant, AE 298 National Defense Education Program — **January 2023 – May 2024**

Aerospace Engineering, University of Illinois Urbana-Champaign

- Taught an introductory **rocketry** course for first- and second-year students, delivering lectures, labs, and build activities.
- Collected, cleaned, and analyzed course/IRB-aligned program data to evaluate student engagement, self-efficacy, and learning impact.
- Redesigned course structure (assessment pacing, project sequencing, and support materials) to improve retention and measurable learning outcomes.

Research Assistant, Grants for Advancement of Teaching in Engineering (GATE) — **November 2023 – May 2024**

Mechanical Engineering, University of Illinois Urbana-Champaign

- Built a **final project framework** using interdisciplinary pedagogy to align hands-on rocketry with course objectives.
- Researched and authored a **literature review** on engineering-education/rocketry; contributed analysis and writing to a conference manuscript.

Teaching Assistant, ASTR 122/210 — **August 2023 – May 2024**

Astronomy, University of Illinois Urbana-Champaign

- Led engaging weekly discussion sections and developed comprehensive exam review materials for over **150** students.
- Created personalized study resources and efficiently graded **150+** assignments and exams, ensuring consistency and timely feedback.

Graduate Career Peer — **August 2022 – May 2024**

Engineering Career Services, University of Illinois Urbana-Champaign

- Advised undergraduate and early graduate students on career planning, resumes/cover letters, and job/internship search strategies.
- Designed and led **mock interview** sessions (behavioral and technical) with targeted, actionable feedback to improve interview performance and confidence.

Graduate Assistant, National Defense Education Program — **January 2023 – May 2023**

Aerospace Engineering, University of Illinois Urbana-Champaign

- Built partnerships with local **K–12** schools to deliver rocketry/avionics outreach, demos, and curriculum-aligned activities.
- Mentored teachers and students on safe build/testing practices and classroom integration; supported impact/data collection to document outcomes.

Calculus, Physics, and Differential Equations Tutor — **January 2019 – May 2022**

Academic Success Center, Iowa State University

- Delivered customized lessons and problem-solving sessions for diverse learners in an inclusive, student-centered environment.
- Drove measurable academic gains, with average grade improvements of **5%–10%** across supported students.